

# Counterfactual Lead Scoring Report

Lead routing report for a provided CSV. The model ranks leads by conversion probability and keeps counterfactual guidance limited to fields a sales or marketing team can reasonably influence.

Rows

1200

Conversion

34.2%

Holdout ROC-AUC

0.934

## Top Lead Drivers

| Feature                         | Importance |
|---------------------------------|------------|
| time_on_site_seconds            | 0.7970     |
| occupation_unknown              | 0.6620     |
| activity_score                  | 0.6504     |
| profile_score                   | 0.2977     |
| lead_origin_lead_add_form       | 0.2320     |
| occupation_working_professional | 0.1388     |
| total_visits                    | 0.0683     |
| page_views_per_visit            | 0.0458     |

## Sample Lead Queue

| Prospect                             | Probability | Tier    | Recommended action                         |
|--------------------------------------|-------------|---------|--------------------------------------------|
| 71abbc0-ea53-4e29-affd-7ab4151a41a8  | 86.2%       | hot     | Route to sales today.                      |
| 8ed7528c-c4a0-4e52-b1b4-a56d63f81154 | 97.3%       | hot     | Route to sales today.                      |
| 2c165c73-6832-4e15-ada3-e91b1c44b9bd | 28.5%       | nurture | Keep in low-touch nurture.                 |
| 0481b169-304f-4814-85bf-65af5381e054 | 53.8%       | warm    | Send a targeted follow-up within 48 hours. |
| 435cb827-db56-4301-9f5e-14e1be5a9ffd | 39.9%       | nurture | Keep in low-touch nurture.                 |
| f8c8282b-33dd-4558-89c1-9eeb1650b433 | 16.5%       | nurture | Keep in low-touch nurture.                 |

## Counterfactual Playbook

- Nurture lead at p=0.29. DiCE counterfactual paths to conversion (behavioral changes only):
- CF 1: time\_on\_site\_seconds: 54.0 -> 1906.0, page\_views\_per\_visit: 6.5 -> 5.2 => probability 0.29 -> 0.78
- CF 2: time\_on\_site\_seconds: 54.0 -> 828.0 => probability 0.29 -> 0.76
- CF 3: time\_on\_site\_seconds: 54.0 -> 1704.0 => probability 0.29 -> 0.81

## Model Risk Notes

- Reported ROC-AUC is measured on a held-out split.
- Do not use post-sale fields, manually assigned sales outcomes, or future activity columns as model inputs.
- Counterfactual guidance should not change immutable or protected customer attributes.